

Dataset 1 – Raw Preamble/IQ Data

- Repository name: RFFI-IQ_only-wifi-802.11g-2.4G-123-m5stack Direct URL to data <https://www.kaggle.com/datasets/yinchen1986/rffi-123-m5stack-iq-wifi-802-11g-2-4g>

RFFI-IQ_only-wifi-802.11g-2.4G-123-m5stack



Data Card Code (2) Discussion (0) Suggestions (0)

About Dataset

This dataset was acquired using 1 NI B-\$210\$ receiver operated via GNU Radio, along with 123 M5Stack Core2 transmitters, 1 Huawei WS700 V2 Wi-Fi access point, and 1 Minisforum NAB6 Lite computer. 1,000 signal frames were collected from each transmitter over a 5-minute interval. All transmitters and the NI B-210 receiver operated based on the IEEE 802.11g standard with a bandwidth of 20\$ MHz. The receiver operated via IEEE 802.11 a/g/p transceiver model with 20 MS/s sampling rate. Specially, each recorded frame mainly contains two elements: (1) MAC address of the device and (2) Raw samples of preamble.

Usability

4.71

License

MIT

Expected update frequency

Not specified

Tags

Data Explorer

952.05 MB

- 24_d7_eb_38_c7_e8_pre.csv
- 24_d7_eb_38_c7_fc_pre.csv
- 24_d7_eb_38_c8_d8_pre.csv
- 24_d7_eb_38_cc_60_pre.csv
- 24_d7_eb_38_cc_e4_pre.csv
- 24_d7_eb_38_cc_f0_pre.csv
- 24_d7_eb_38_cd_00_pre.csv
- 24_d7_eb_38_cd_5c_pre.csv
- 24_d7_eb_38_cf_ec_pre.csv
- 24_d7_eb_38_d0_b0_pre.csv
- 24_d7_eb_38_d0_cc_pre.csv
- 24_d7_eb_38_d6_c4_pre.csv
- 24_d7_eb_38_d8_64_pre.csv
- 24_d7_eb_38_da_68_pre.csv
- 24_d7_eb_38_de_1c_pre.csv
- 24_d7_eb_38_de_dc_pre.csv
- 24_d7_eb_38_e0_a4_pre.csv
- 24_d7_eb_38_e0_dc_pre.csv
- 24_d7_eb_38_e1_04_pre.csv
- 24_d7_eb_38_e1_08_pre.csv

24_d7_eb_38_c7_e8_pre.csv (7.86 MB)

Detail Compact Column

3 of 3 columns

Suggest Edits

# Device Number	MAC_address	preamble
1	1	1000
1	24:d7:eb:38:c7:e8	[0.21494181+1.1249119 8e-01j -0.18778035+5.978574 60e-02j -0.19699693- 3.15866508e-02j -0.04773094+3...
1	24:d7:eb:38:c7:e8	[6.95821829e-

Dataset 2 - Extracted RF Features + IQ Data

- Repository name: RFFI-kf_feature-IQ-wifi-802.11g-2.4G-123-m5stack Direct URL to data <https://www.kaggle.com/datasets/yinchen1986/rffi-kf-feature-iq-wifi-802-11g-2-4g-123-m5stack>

MAC_address	preamble	iq_preamble	CFO	short_freq	long_freq	frac_dimension_1	frac_dimension_2	iqi_1	iqi_2	mag_error_vec_1	mag_error_mean_1	mag_error_var_1	mag_error_vec_2	mag_error_mean_2	mag_error_var_2	phase_error_vec_1	phase_error_mean_1	phase_error_var_1	phase_error_vec_2	phase_error_mean_2	phase_error_var_2
24:d7:eb:38:c7:e8	[0.2149418 1+1.12491 198e-01j - 0.1877803 5+5.97857 460e-02j - 0.051797+0	[0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0.056486- 0.0012075i, 0.056201+0 .0067885i, 0.051797+0	-13449	0.0076333	-0.01186	1.94521183	1.94521183	3.1	3.1	[0, 0, 0, 0, 0, 0, 0.9435, 0.94339, 0.93088, 0.94781, 0.93304, 0.94134,	0.7096838	0.1214105	[0, 0, 0, 0, 0, 0, 0.9435, 0.94339, 0.93088, 0.94781, 0.93304, 0.94134,	0.7096838	0.1214105	[0, 0, 0, 0, 0, 0, 0.021373, -0.12021, 2.4179, 2.3975, - 0.51497, - 0.47713, 2.4487, -	1.01628851	2.60134485	[0, 0, 0, 0, 0, 0, -3.1202, 3.0214, 5.5595, 5.539, 2.6266, 2.6645, 5.5903,	1.21263805	6.77103655
24:d7:eb:38:c7:e8	[6.9582182 9e- 03+0.0980 8647j - 1.1072112 6e-01- 0.0075075	[0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0.0001158- 0.015442i, 0.010314- 0.025965i, 0.0011546-	-14333	0.0059857	-0.01049	1.98460317	1.98460317	6.82	6.82	[0, 0, 0, 0, 0, 0, 0.98456, 0.97206, 0.97037, 0.97565, 0.98116, 0.98059,	0.7711075	0.1403295	[0, 0, 0, 0, 0, 0, 0.98456, 0.97206, 0.97037, 0.97565, 0.98116, 0.98059,	0.7711075	0.1403295	[0, 0, 0, 0, 0, 0, 1.5633, 1.1927, 4.6734, 4.4511, 1.943, 1.8193, 2.1477,	1.25222033	2.99540827	[0, 0, 0, 0, 0, 0, -1.5783, -1.9489, 1.5318, 1.3095, - 1.1986, - 1.3223, 5.2893, -	0.17229786	6.54503861
24:d7:eb:38:c7:e8	[0.1992858 2- 1.1496319 6e-01j - 0.0513321 2+2.00689 673e-01i	[0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0+0i, 0.049884+0 .017883i, 0.042601+0 .013117i, 0.047344+0	-16640	0.0071055	-0.01233	1.94990715	1.94990715	2.57	2.57	[0, 0, 0, 0, 0, 0, 0.94701, 0.95542, 0.94766, 0.93026, 0.9381, 0.95667,	0.7104552	0.1211758	[0, 0, 0, 0, 0, 0, 0.94701, 0.95542, 0.94766, 0.93026, 0.9381, 0.95667,	0.7104552	0.1211758	[0, 0, 0, 0, 0, 0, -0.34421, -0.2987, 2.7011, 2.8285, - 0.37563, - 0.22416, 2.4299, -	1.00703256	2.68745317	[0, 0, 0, 0, 0, 0, 2.7974, 2.8429, 5.8427, 5.9701, 2.766, 2.9174, 5.5715,	1.30155687	6.05799301